

ZaraiLink — Speaker Script

What to actually say for slides 4, 5, 6

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Total speaking budget: 10 minutes

How to read this. What's in quotes is roughly what you say. Square brackets like **[45s]** are timing cues. Italic notes like *point at the diagram* are stage directions. Don't memorise word-for-word — catch the rhythm and improvise.

1 Slide 4 — System Architecture

Skim before the talk — `zarailink_guide.pdf`

Priority	Pages	Why
Must	15–16	The layered diagram on page 15 mirrors the slide; §3.2 traces a request end-to-end.
Should	17	§3.3 auth flow + §3.4 token economy.
Should	18	§3.5 caching strategy + §3.6 data lineage.
Optional	1–3	§1.2 elevator pitch if you want a one-line opener.

Script (≈ 3 minutes)

[15s] *point at the slide.* “Okay so this is the architecture. Three pieces I want to walk you through: the user side, the backend, and the intelligence layer that lives inside it.”

[45s] *walk left to right across the diagram.* “On the left, that's the user opening our web app in a browser. The browser talks to our backend over a REST API — queries go in as JSON, results come back as JSON. The backend talks to a PostgreSQL database, and that's where everything lives: users, companies, transactions, the whole product catalog.”

[60s] *point at the intelligence layer — this is the part the audience cares about.* “Now this middle box is the intelligence layer — this is the part that makes the platform actually feel intelligent. When a search query comes in, it flows through four stages.”

“Stage one is NLU. We figure out what the user wants — are they buying or selling, what product, which country, what price range. Two ML models do this: SetFit for intent, GLiNER for pulling product names and countries out of messy queries.”

“Stage two is candidate retrieval. We map the query to the right product sub-categories using keyword search, semantic embeddings, and fuzzy matching for typos.”

“Stage three is aggregation. One heavy Postgres query rolls up every transaction per supplier — average price, total volume, shipment frequency, recency.”

“Stage four is ranking. We score every candidate with a seventy-thirty blend of hand-tuned heuristics and a LightGBM ranker.”

[30s] *the warmup — this is a fun detail to drop.* “One thing worth knowing — when the server boots, we run two throwaway searches in a background thread before any real user shows up. That preloads all the ML models into memory. Without it, the first search of the day would take five seconds while everything cold-starts. With it, the first query and the thousandth query both come back in milliseconds.”

[30s] *point at Redis and OpenRouter.* “Off to the side — Redis is our cache. If it’s running, we share parsed query results across requests so identical queries don’t redo all the ML work. If Redis is down, we fall back to an in-memory cache automatically. And this OpenRouter box — we only call out to a language model when our regex price extraction fails on something tricky like “cheap basmati under forty cents”. About one query in twenty actually hits it. The rest stay completely local.”

2 Slide 5 — Use Case and Dataset

Skim before the talk — `zarailink_guide.pdf`

Priority	Pages	Why
Must	64–65	§7.1 spreadsheet specs — exact row counts and columns.
Must	4	§1.6 user-facing tour — the seven pages map to the use cases.
Should	32	§4.5.1 the four-level product hierarchy.
Should	67–68	§7.5 ingestion scripts — how raw rows become DB rows.

Script (≈ 3 minutes)

[60s] *point at the use-case diagram.* “Two kinds of users on the platform. Regular users — so that’s importers, exporters, analysts — and admins.”

“Regular users can do five things. They can search, and we offer four ways to search: by HS code, by category, by product name, or by typing a question in plain English. They can view a market-intelligence dashboard with charts. They can browse the company directory. They can unlock premium contact information using tokens — discovery is free, but pulling up a phone number or email spends tokens from their balance. And they can sign in and manage their account.”

“Admins do two things — they manage user accounts, and they ingest and curate the trade data itself.”

[90s] *this is the part that needs concrete numbers — slow down and hit each one.* “Now, the data. This is what makes the whole thing real instead of a toy.”

“Our industry partner is PAR. They handed us roughly nine thousand raw transaction records — sugar imports and exports spanning 2024 to 2025. Out of those nine thousand, we sat down and manually cleaned and standardized about five thousand. That meant fixing column headers, formatting HS codes to a uniform four-decimal style, and fuzzy-matching company names that were spelled three different ways across different rows.”

“Specifically: around 2,200 import transactions covering twelve months ending November 2024, and around 7,200 export transactions on the same schema — just with the buyer and seller

flipped, because Pakistan is now the seller.”

[30s] *end on the hierarchy — this connects to the ERD slide. “All of it goes into Postgres in a four-level hierarchy: Product, then Category, then Sub-Category, then Item. We didn’t invent this — it mirrors how the World Customs Organisation actually structures HS codes. Modeling our database the same way means analytics can roll up to whichever level the user is asking about, without us having to maintain separate tables for separate granularities.”*

3 Slide 6 — ERD

Skim before the talk — [zarailink_guide.pdf](#)

Priority	Pages	Why
Must	88–90	Appendix C — model reference. Every table on the ERD with its key fields.
Must	32–33	§4.5.1–4.5.3 + Table 4.7 — Transaction model and product hierarchy.
Should	17	§3.4 token economy — explains the unlock tables.

Script (≈ 3 minutes)

Strategy. Don’t read field names off the slide. The audience can see the ERD — give them the four-cluster shape and one interesting design choice, then move on.

[15s] *open with reassurance — the audience is staring at a wall of tables. “This looks like a lot, but it’s really just four groups of tables, and each group matches one part of the application.”*

[20s] *cluster 1 — keep brief. [accounts_user] “First group is identity. Just the user table. Email is the login field, and every user has a token balance integer right on their row — that’s the in-app currency for unlocking premium data.”*

[45s] *cluster 2 — the heart of the schema, this is where to spend the time. [trade_data_product, trade_data_productcategory, trade_data_productsubcategory, trade_data_productitem, trade_data_transaction, trade_data_productembedding] “Second group is the catalog. The four-level product hierarchy — Product, Category, Sub-Category, Item — each level pointing to its parent. And then the transaction table, which is the ledger every other table builds on. Every row is one shipment: when, what HS code, who bought, who sold, where from, where to, how much, at what price. The search engine spends most of its time querying this table, so we’ve indexed every column the search filters by — which is what lets a query return in milliseconds even on the full dataset.”*

[20s] *cluster 3 — brief. [companies_company, companies_sector, companies_keycontact, companies_keycontactunlock] “Third group is the directory — companies, their sector, their key contacts, and a small junction table that records which user has unlocked which contact.”*

[30s] *cluster 4 — end with the one design-choice point worth volunteering. [subscriptions_subscriptionplan, subscriptions_usersubscription, subscriptions_useraccess] “Fourth group is the access side — subscription plans, user subscriptions, and a record of what each user has unlocked at the product level. You’ll notice we have two separate unlock tables — one for individual contacts*

(over in the directory cluster), one for whole product categories (here). They look like duplicates but they aren't — different granularities of access, different lookups, so they're kept separate."

[10s] closer. *"So that's the schema in one breath: identity, catalog, directory, access. Each cluster is its own Django app, so changes in one don't break the others."*

Time check

Slide	Budget	Where to spend the time
Slide 4	3 min	The diagram walk + the "why we embedded the ML" moment.
Slide 5	3 min	Use cases brisk. Dataset numbers slow — those are the academic credibility.
Slide 6	~2 min	Four clusters at a brisk pace. Audience already sees the ERD — give them shape, not field lists.
Buffer	~1 min	Transitions and audience questions.

One last thing. Practise once out loud with a stopwatch. The first time through is always 30 seconds longer than you planned. After that you'll know which sentences to drop if you're running long.